

Solar panel manufacture is an area which has seen significant growth in recent years—Nathalie Morin, marketing communications manager, OK International



OK International's MFR series desoldering system

es the static charge of the user's body to ground every time he picks up the soldering iron.

Non-ESD-safe stations are those which do not guarantee any static charge safety. In these stations, the ESD would transfer to the soldering tip, thus destroying the IC being soldered.

"ESD-safe stations are required for PCBs that contain sensitive components prone to damage by electrostatic charges. For normal PCBs that do not contain such components, non-ESD-safe stations suffice the purpose," shares Jain.

Lead-free and non-lead-free

"Lead-free soldering means soldering without the use of lead—a harmful element for the environment. Normally used tin-lead solders, also called 'soft solders,' are commercially available with tin concentrations between 5 and 70 per cent by weight. The greater the tin concentration, the greater the sol-

der's tensile and shear strengths," explains Chadda.

At the retail level, the two most common alloys are 60/40 tin/lead (Sn/Pb), which melts at 370°F or 188°C, and 63/37 Sn/Pb, which is used principally in electrical/electronic work. The health and liability risks associated with the use of lead have led to government regulations concerning the elimination of lead from certain manufacturing processes.

Lead has already been banned from paint, plumbing and gasoline. Solder too is likely to be regulated in the near future.

Introducing lead-free soldering generally involves process changes as well as equipment modifications, as lead-free alloys have a significantly higher tin content than tin-lead solder, and require higher processing temperatures. However, a growing number of users are currently updating their soldering equipment to lead-free because a lead-free pot will also accommodate common tin-lead alloys.

"Lead-free soldering has arisen because of legislation being enacted around the globe. In the EC, the WEEE (Waste from Electrical and Electronics Equipment) Directive has brought a sharp focus onto lead-free soldering technology. Although this directive is mainly about recycling, it also contains clauses aimed at banning the use of lead in certain categories of electrical and electronic equipment. This gives

rise to the need for lead-free soldering," informs Chadda.

"Lead-free solder has a higher melting point. So stations need to be capable of fast heat generation and recovery," says Gupta.

One type of solder that is gaining acceptance is based on an alloy of tin, silver and copper. This family of solder offers a high reliability and good solderability, but it has a slightly higher melting point (217°C) than the traditional tin-lead solder. Also, this lead-free solder is more costly because of the silver content.

For applications where a lower melting point is needed, another combination using tin, zinc and bismuth offers a melting point of just 195°C. This is particularly useful for those applications where damage may be caused to the components or the boards if the temperature is raised too far. Another alloy using tin, silver and bismuth also offers a low melting point.

Lead-free solder containing tin with just 0.7 per cent copper is another possibility. This type of solder can be obtained in wire format for manual soldering.

In terms of performance, the tin-silver-copper alloy is able to provide similar levels of reliability to the traditional tin-lead alloys, but to achieve this the process has to be implemented correctly, and often higher temperatures may be needed.

Lead-free products are gaining ground. Blue Sky Systems provides its customers with the option of lead-free soldering stations and other lead-free solder products to meet their requirements.

Variable-power soldering

Whilst conventional soldering systems have improved in recent years, these are still fixed-power systems that store energy in their tip, and don't adjust power in response to the thermal demand. The result is that the tip cools as it heats the joint and the heater reheats the tip, often over-

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